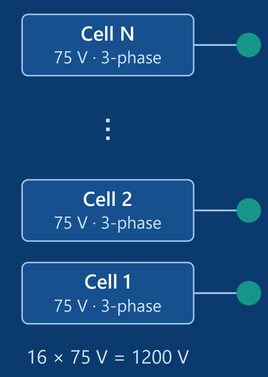
Moore4Power General Assembly · November 2026

Stacked Polyphase Bridge Traction Drives: Iterative GaN Building Block Design

Enes Ayaz, Shahriar Sarmast Ghahfarokhi, Kevin Klarin, Staffan Norrga

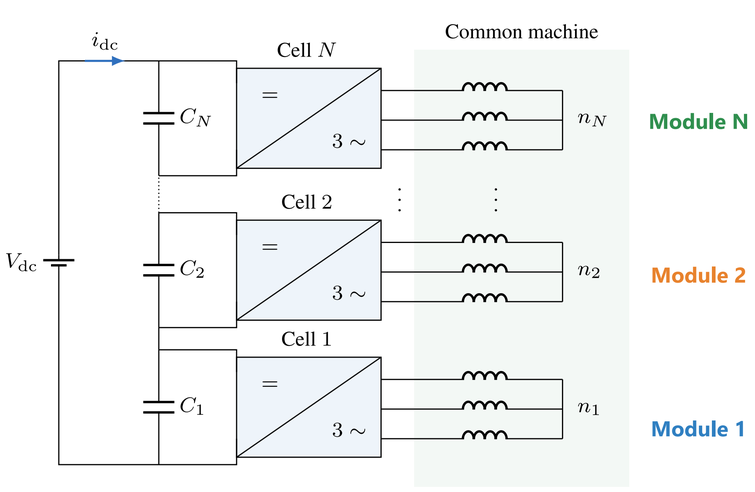
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1200 V	16 × 75 V	4 × GaN	0.41 nH	99.71 %	2.85 %
dc link, 300 kW traction	series cells, own winding groups	EPC2361 100 V devices per switch	commutation loop per cell	semiconductor efficiency	shared-bus dc ripple, 50 kHz

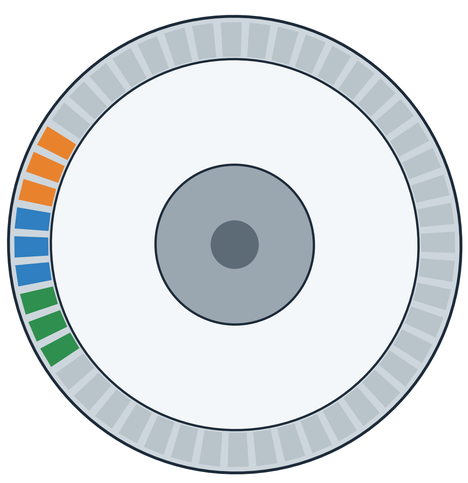
1 Concept: split the dc voltage, keep one machine

N-cell SPB

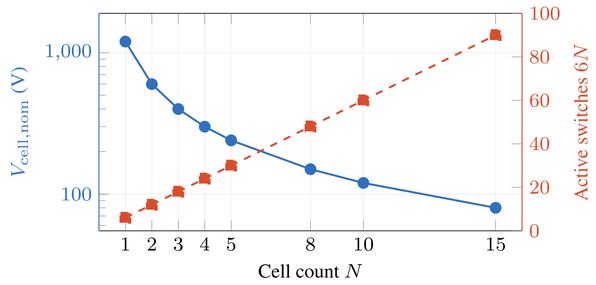


Series dc links, one three-phase winding group per cell

Machine: N isolated winding sets



Device voltage class and cell count



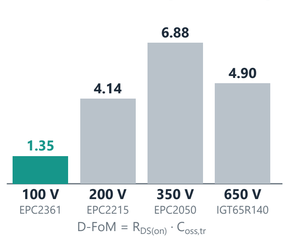
N	1	2	3	4	5	8	10	15
V_{cell} (V)	1200	600	400	300	240	150	120	80
$6N$	6	12	18	24	30	48	60	90

1200 V link: $V_{cell} = V_{dc}/N$ versus active switches 6N

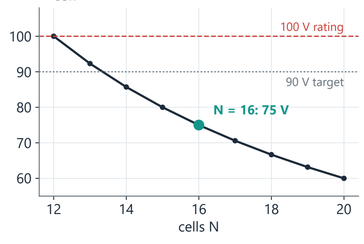
- N two-level bridges in series on the dc side; each feeds its own three-phase winding group.
- $V_{dc,k} \approx V_{dc}/N$: the device voltage class follows the cell, not the full link.
- All cells carry the same dc current; $P_{ac} = \Sigma P_k$, so cell voltages must be balanced actively.
- 6N switches, N drivers and capacitors: complexity moves to the system.
- Rating with margin: $V_{rated} \geq k_{rel}(V_{dc,max}/N + \Delta V_{imb} + \Delta V_{sw})$.
- N = 16 gives 75 V cells for 100 V GaN; the overshoot L·di/dt sets the usable margin.

2 Voltage class, cell count, single cell

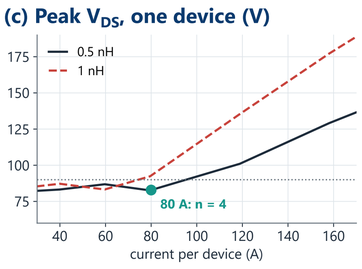
(a) Lowest D-FoM per class (ps)



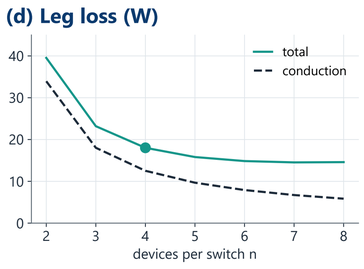
(b) V_{cell} = 1200 V / N



(c) Peak V_{DS}, one device (V)



(d) Leg loss (W)

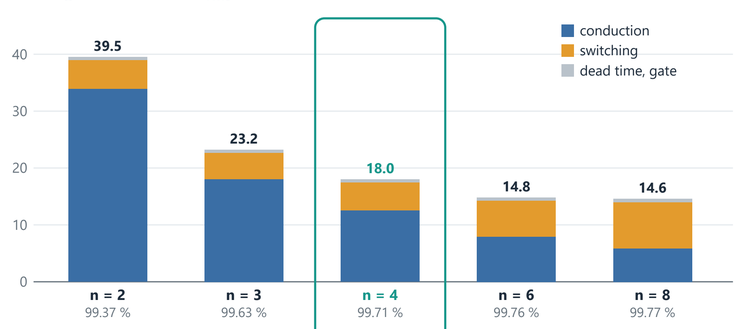


Lower blocking voltage gives the best figure of merit (D-FoM).

Four devices: 18 W per leg, the knee of the loss curve

3 Why four devices per switch

(a) Leg loss at 228 A_{rms}, 50 kHz (W)



(b) Screens

	n = 2	n = 3	n = 4	n = 6	n = 8
T_j	✗ 141 °C	✓ 84 °C	✓ 69 °C	✓ 59 °C	✓ 56 °C
V_{DS}	✗ 131 V	✗ 96 V	✓ 83 V	✓ 86 V	✓ 83 V

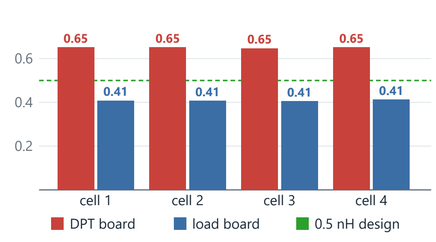
Limits: $T_j \leq 125$ °C, $V_{DS} \leq 90$ V at 0.5 nH.

n = 4 is the smallest count that passes both.

- n = 4: T_j 69 °C, 2.24 W per device, 8 driver channels per leg
- 1 nH instead of 0.5 nH: 93 V peak, still below the rating

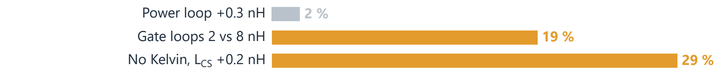
4 Extracted parasitics (Ansys Q3D)

Commutation loop per cell (nH)

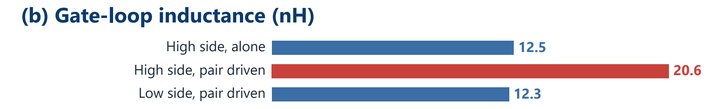


- **0.41 nH** load board, 0.65 nH DPT (shunt)
- Cells within 1.6 %
- Cell coupling k ≤ 0.034

(a) Switching-energy imbalance, two devices

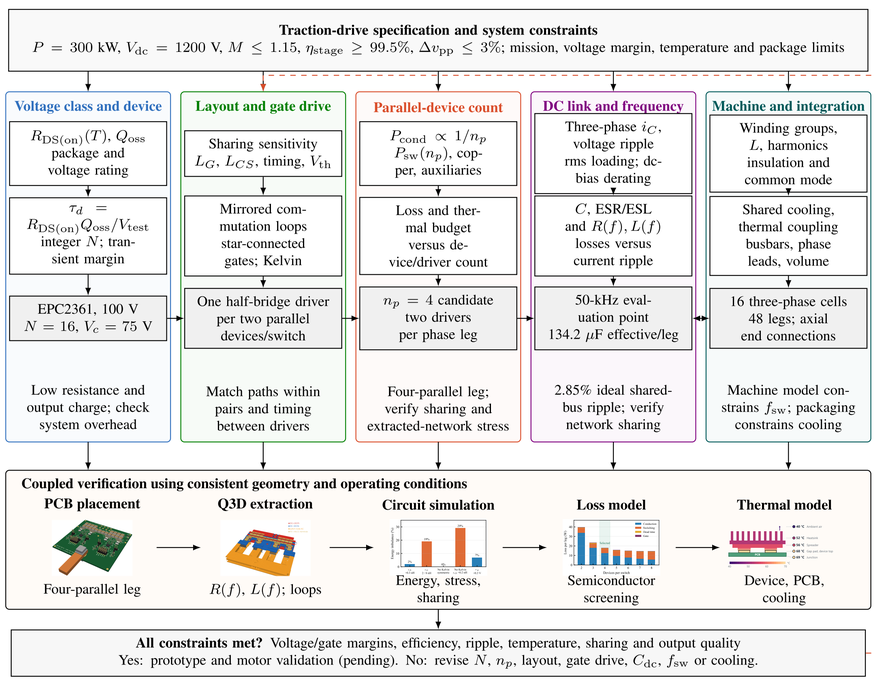


(b) Gate-loop inductance (nH)



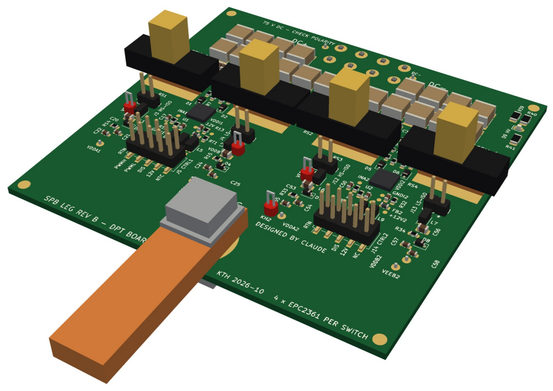
One 2EDF7275K channel per device pair, star gate network with Kelvin returns

5 Iterative system design (review)

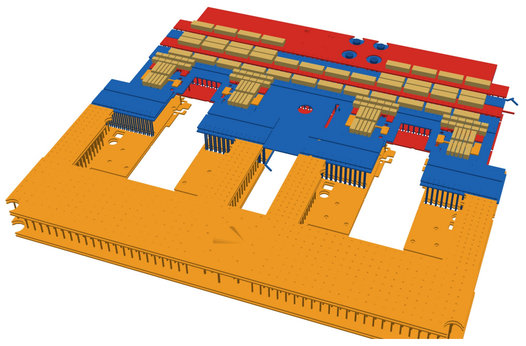


Constraints → device, layout, parallel count, dc link, machine → coupled verification

6 Building block: double-pulse-test board and extracted network



DPT board: four cells, eight EPC2361



Ansys Q3D model of the four cells

0.41 / 0.65 nH
loop per cell, load / DPT

20.6 / 12.3 nH
gate loop, HS / LS pair

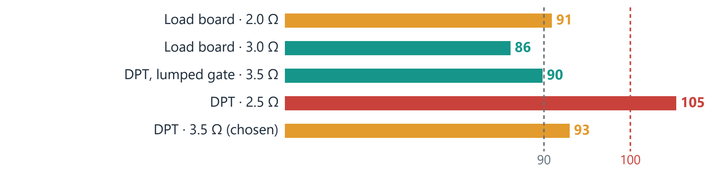
DC+
DC−
switch node
FET / MLCC
Q3D model colours

LTspice with the extracted network

75 V, 322 A, four cells, extracted gate loops

- High-side v_{DS} peak **93 V** < 100 V rating at $R_{g,on}$ 3.5 Ω
- Four parallel high-side gate loops equal within 0.3 %
- Turn-on energy shared within 0.2 %

Peak high-side V_{DS} (V)



axis from 60 V; dashed: 90 V target, 100 V rating

✓ Conclusions

1. **16 × 75 V cells let 100 V GaN serve a 1200 V, 300 kW drive**
   The 100 V class has the lowest D-FoM; EPC2361 is the device.
2. **Four devices per switch: 18 W per leg (99.71 %)**
   69 °C junction and 83 V peak at 0.5 nH loop inductance.
3. **The extracted gate loop (20.6 nH) sets the margin**
   3.5 Ω turn-on and −1.9 V off-state bias keep the peak at 93 V.

7 Project status and next steps

- DPT board: waiting for PCB manufacturing and assembly (JLCPCB).
- Double-pulse tests at 75 V up to 322 A: per-cell current sharing, overshoot, E_{on}/E_{off}, gate bump.
- Then three load-test legs + PCB busbar: one 75 V three-phase level, continuous and thermal tests.

8 Open research questions

- Optimum N: device class versus component count shifts with SiC, GaN and cooling.
- Joint battery–converter–machine design on vehicle energy, mass, cost and reliability.
- Stack-position common-mode voltage, bearing currents and insulation across the full stack.